

Mitigation strategies to reduce enteric methane emissions from dairy cows: Update review

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Boadi, D., Benchaar, C., Chiquette, J. and Massé, D. 2004. **Mitigation strategies to reduce enteric methane emissions from dairy cows: Update review.** *Can. J. Anim. Sci.* **84**: 319–335. Enteric methane (CH₄) emission is a major contributor to Canadian greenhouse gas emissions, and also a loss of feed energy during production. The objective of this paper is to provide an update on current management practices and new dietary strategies recently proposed to reduce CH₄ emissions from ruminants. Existing mitigation strategies for dairy, e.g., the addition of ionophores, fats, use of high-quality forages, and increased use of grains, have been well researched and applied. These nutritional changes reduce CH₄ emissions by manipulating ruminal fermentation, directly inhibiting methanogens and protozoa, or by diverting hydrogen ions away from methanogens. Current literature has identified new CH₄ mitigation options. These include the addition of probiotics, acetogens, bacteriocins, archaeal viruses, organic acids, plant extracts (e.g., essential oils) to the diet, as well as immunization, and genetic selection of cows. These new strategies are promising, but more research is needed to validate these approaches and to assess in vivo their effectiveness in reducing CH₄ production by dairy cows. It is also important to evaluate CH₄ mitigation strategies in terms of the total greenhouse gas budget and to consider the cost associated with the various strategies. More basic understanding of the natural differences in digestion efficiencies among animals as well as a better knowledge of methanogens and their interaction with other organisms in the rumen would enable us to exploit the potential of some of the new CH₄ mitigation strategies for dairy cattle production.

Key words: Enteric methane, dairy cattle, mitigation

Boadi, D., Benchaar, C., Chiquette, J. et Massé, D. 2004. **Stratégies visant à réduire les dégagements de méthane entérique des vaches laitières: bilan.** *Can. J. Anim. Sci.* **84**: 319–335. Le méthane (CH₄) d'origine entérique contribue dans une large mesure aux émissions canadiennes de gaz à effet de serre et entraîne une perte de l'énergie tirée des aliments durant la production. Le présent article fait le point sur les pratiques de zootechnie actuelles et sur les stratégies d'engraissement récemment proposées pour réduire les dégagements de CH₄ des ruminants. Les stratégies d'atténuation existantes (addition d'ionophores, matière grasse, utilisation de fourrages de grande qualité, plus grande utilisation de grains) ont fait l'objet de maintes recherches et sont désormais bien appliquées. Les changements nutritionnels qu'elles introduisent diminuent les émissions de CH₄ en modifiant la fermentation dans le rumen, en inhibant directement la production de méthanogènes et la population de protozoaires ainsi qu'en détournant les ions hydrogène des méthanogènes. La documentation scientifique mentionne de nouvelles méthodes pour atténuer la libération de CH₄. Ces méthodes comprennent l'ajout de probiotiques, d'acétogènes, de bactériocine, de virus archaïques, d'acides organiques et d'extraits végétaux (huiles essentielles) à la ration des animaux de même que l'immunisation et la sélection génétique des vaches. Ces nouvelles stratégies semblent prometteuses, mais elles doivent faire l'objet de recherches plus poussées si on veut les valider et évaluer in vivo l'efficacité avec laquelle elles réduisent la production de CH₄ par les vaches laitières. Les stratégies d'atténuation doivent aussi être évaluées d'après la production globale de gaz à effet de serre et on doit tenir compte des coûts qui s'y associent. On devrait parvenir à une meilleure compréhension des différences naturelles affectant la digestion des animaux ainsi qu'acquérir des connaissances sur les méthanogènes et la façon dont ils interagissent avec d'autres unicellulaires du rumen en vue de mieux exploiter le potentiel de certaines nouvelles stratégies d'atténuation au niveau de l'élevage des bovins laitiers.

Mots clés: Méthane entérique, bovins laitiers, atténuation

It has been well documented that the accumulation of greenhouse gases (GHG), mainly carbon dioxide (CO₂), methane (CH₄) and nitrous oxide (N₂O), in the atmosphere is contributing to an increasing earth surface temperature [Intergovernmental Panel on Climatic Change (IPCC) 1994; Cole et al. 1997; Moss et al. 2000]. The accumulation of these gases is known to be increasing at rates from 0.3 to 0.9% per year, largely because of anthropogenic effects on the carbon and nitrogen cycles (Janzen et al. 1999; Desjardins et al. 2001). The IPCC has asked nations to quantify the amount of gases they produce and to develop

research to limit further emissions (Moss et al. 2000). In order to compare emissions of GHG, the global warming potential (GWP) of the individual gases is used, with CO₂ as the reference gas. The GWP of CH₄ is known to be 21-fold greater than that of CO₂ and the GWP of N₂O is 310-times greater than that of CO₂ (Ruminant Livestock Efficiency Program 1997). Thus, when the emissions of CH₄ and N₂O

Abbreviations: BES, bromoethanesulphonate; BW, body weight; DM, dry matter; DMI, dry matter intake; GEI, gross energy intake; GHG, greenhouse gas; GWP, global warming potential; IPCC, International Panel on Climate Change; RFI, residual (net) feed intake; TMR, total mixed ration; VFA, volatile fatty acids

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are multiplied by their GWP, the emissions are expressed as CO₂-equivalents (Janzen et al. 1999; Desjardins et al. 2001).

Canada's GHG emissions from anthropogenic sources were approximately 599 Tg CO₂-equivalents in 1990 (Jacques et al. 1997; Desjardins et al. 2001). In accordance with the Kyoto protocol in 1997, Canada agreed to lower its emissions to 6% below its 1990 levels by 2008–2012. To achieve this, Canada is expected to lower GHG emissions to approximately 563 Tg CO₂-equivalents by 2008–2012 (Desjardins et al. 2001). Greenhouse gases from all agricultural sources, which include the use of fossil fuels, were 90.5 Tg CO₂-equivalents in 1991 and estimated to be 98.0 Tg CO₂-equivalents in 2010, under a business as usual scenario, which assumes that the present trends in management practices will continue (Desjardins et al. 2001). For the agriculture sector, emissions must therefore be reduced by about 12.9 Tg CO₂-equivalents from projected emissions by 2010, under a "business as usual" scenario (Desjardins et al. 2001). The emissions from livestock production are estimated to be 35.6 Tg CO₂-equivalents in 2010, compared to 30.3 Tg CO₂-equivalents in 1991 (Desjardins et al. 2001). Therefore emissions by the livestock sector must be reduced by 7.1 Tg CO₂-equivalents from projected emissions by 2010. Accordingly, different strategies have been proposed to reduce enteric CH₄ emissions from ruminants.

Approximately 10% of the Canadian GHG emissions are attributed to agricultural activities, not including the use of fossil fuels or the indirect GHG emissions from fertilizer production [Agriculture and Agri-Food Climate Change (AAFC) Table Options Report 2000]. Nitrous oxide emissions associated with agricultural nitrogen sources (fertilizer and animal manure) represent 61% of agricultural GHG emissions. Methane represents 38%, while net CO₂ accounts for less than 1% of GHG emissions (AAFC 2000). Livestock production systems contribute about 42% of total GHG production from agriculture with 28% associated with direct emissions of enteric fermentation and 14% from indirect emissions from livestock operations, related to manure handling, storage, and land application (AAFC 2000). Canadian cattle account for 97% of total livestock enteric CH₄ emissions; 25% is attributed to dairy and beef cattle account for 72% (Janzen et al. 1999). Canadian livestock enteric CH₄ emissions were 18 Mt CO₂-equivalents in 1996 and have increased by 14% since 1990, due to an increased number of heads (Neitzert et al. 1999).

Different approaches have been proposed to reduce CH₄ production by ruminants. The aim of this paper is to review some of the current management practices available for mitigation and new strategies proposed to mitigate enteric CH₄ emissions from ruminants, as they relate in particular to dairy cattle.

ESTIMATION OF ENTERIC METHANE EMISSION

Currently, CH₄ emissions from enteric fermentation for Canadian cattle are estimated by multiplying the population of various classes of animals by average emission factors derived for each type of domestic animal (Table 1), which are set by the guidelines of IPCC (Neitzert et al. 1999). The

IPCC CH₄ emission values are based upon prediction equations and models, which are themselves based on the following relationship between CH₄ production, feed intake and digestibility (Blaxter and Clapperton 1965):

$$\text{CH}_4 (\% \text{ of GEI}) = 1.3 + 0.112 D + L (2.37 - 0.05D)$$

where GEI = gross energy intake, L = level of feed intake and D = dry matter digestibility. The Blaxter and Clapperton (1965) prediction equation was developed from respiration calorimetry chamber experiments using mainly sheep, and is best suited for estimating CH₄ emissions when feed types and feeding levels are the same as those used to develop the model. The equation above predicts emission loss in the range of 5 to 8% of GEI. However, observed CH₄ emissions from a wide range of feeds and animals varied from 2 to 12% of GEI (Johnson and Johnson 1995). Using an extensive database ($n = 452$), Johnson and Johnson (1995) showed that the ability of the Blaxter and Clapperton's equation to predict CH₄ emissions was weak; i.e., the relationship between predicted and observed CH₄ emissions was very poor ($r^2 = 0.23$).

The literature also provides evidence that enteric fermentation can vary widely depending on factors such as type of the animal, the amount and type of feed, environment, addition of dietary fat, feed additives and body weight of the animal (Mathison et al. 1998; Moss et al. 2000). Therefore, IPCC data (1994) may over or under estimate emissions produced by Canadian cattle production systems where animals are under different feeding and environmental conditions from those under which IPCC data were derived.

METHANE PRODUCTION IN THE RUMEN

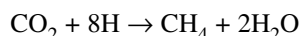
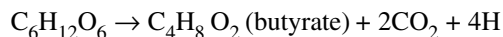
Methanogenesis

Enteric CH₄ emission is produced as a result of microbial fermentation of feed components. Methane, a colorless, odorless gas, is produced predominantly in the rumen (87%) and to a small extent (13%) in the large intestines (Murray et al. 1976; Torrent and Johnson 1994). Rumen CH₄ is primarily emitted from the animal by eructation. The conversion of feed material to CH₄ in the rumen involves the integrated activities of different microbial species, with the final step carried out by methanogenic bacteria (McAllister et al. 1996; Moss et al. 2000). Primary digestive microorganisms (bacteria, protozoa and fungi) hydrolyze proteins, starch and plant cell wall polymers into amino acids and sugars. These simple products are then fermented to volatile fatty acids (VFA), hydrogen (H₂), and CO₂ by both primary and secondary digestive microorganisms. Acetate, propionate, and butyrate, which are the major VFA, are then absorbed and utilized by the host animal. The major producers of H₂ are the organisms which produce acetic acid in the fermentation pathway (Van Soest 1982; Hegarty and Gerdes 1998).



Table 1. Canadian cattle enteric methane emission factors used by IPCC^z

Species	(L CH ₄ d ⁻¹) ^y	(kg CH ₄ yr ⁻¹) ^y
Bulls	288	75
Dairy cows	453	118
Beef cows	276	72
Dairy heifers	215	56
Beef heifers	215	56
Heifers (slaughter)	180	47
Steers	180	47
Calves	180	47

^zAdapted from Neitzert et al. (1999).^yCalculated figures.

Although H₂ is one of the major end products of fermentation by protozoa, fungi and bacteria, it does not accumulate in the rumen. It is used by other bacteria, mainly the methanogens which are present in the mixed microbial ecosystem. Moss et al. (2000) established that CH₄ production can be calculated from the stoichiometry of the main VFA formed during fermentation, i.e., acetate (C₂), propionate (C₃) and butyrate (C₄) as follows: CH₄ = 0.45 C₂ – 0.275 C₃ + 0.40 C₄. Thus, the molar percentage of VFA influences the production of CH₄. Acetate and butyrate production results in CH₄ production, while propionate formation serves as a competitive pathway for H₂ use in the rumen. With an increased molar proportion of propionate, the molar proportions of acetate and/or butyrate are reduced (Baker 1997).

Methanogens

Methanogens represent a unique group of microorganisms. They possess three coenzymes which have not been found in other microorganisms. The three coenzymes are: coenzyme 420, involved in electron transfer in place of ferredoxin, coenzyme M, involved in methyl transfer, and factor B, a low molecular weight, oxygen-sensitive, heat-stable coenzyme involved in the enzymatic formation of CH₄ from methyl coenzyme M (Jones et al. 1987; Baker 1999). Methanogens in all habitats differ from almost all bacteria in cell envelope composition: there is no muramic acid in the cell wall, and the cell membrane lipids are composed of isoprenoids ether-linked to glycerol or other carbohydrates (Baker 1999). Analyses of the nucleotide sequence of the 16S ribosomal RNA indicate their very early evolutionary divergence from all other forms of life studied so far. They have therefore been classified in a different domain named the Archae (formerly Archaeobacteria) within the kingdom Euryarchaeota (Baker 1997, 1999). Methanogens are nutritionally fastidious anaerobes and grow only in environments with a redox potential below –300 mV (Stewart and Bryant 1988). Most methanogens grow at neutral pH, between 6 and 8. However some species can thrive in environments with pH extremes from 3 to 9.2 (Jones et al. 1987). Five species of methanogens were reported to have been isolated in the rumen (McAllister et al. 1996). These include

Methanobrevibacter ruminantium, *Methanosarcina barkeri*, *Methanosarcina mazei*, *Methanobacterium formicicum* and *Methanomicrobium mobile*. Only *Methanobrevibacter ruminantium* and *Methanosarcina barkeri* have been found in the rumen at populations greater than 10⁶ mL⁻¹, and are assumed to play a major role in ruminal methanogenesis (Lovley et al. 1984). In recent years, phylogenetic analysis of Archaeal 16S rRNA genes cloned from the rumen showed that most of the organisms present differed from the cultivated species (Whitford et al. 2001). It has been suggested that there may still be more methanogens not yet identified, and more will be identified as 16S rRNA analysis progresses.

Methanogens use the process of formation of CH₄ to generate energy for growth. Substrates used in the process include H₂, CO₂, formate, acetate, methanol, methylamines, dimethyl sulfide, and some alcohols (Jones 1991; McAllister et al. 1996). In the rumen, methanogens primarily use H₂, CO₂ and formate as substrates in methanogenesis (Jones 1991). The unique biochemical ability of *Methanosarcina barkeri* to use methanol, methylamines, and acetate in addition to CO₂ and H₂ as substrates enables the slow growing *Methanosarcina* organisms to flourish in ruminants fed diets containing ingredients like molasses that break down into methylamines, methanol and acetate (Vicini et al. 1987). Only two species (*Methanosarcina* and *Methanosaeta*) are known to degrade acetate to CH₄ in the rumen (Jones 1991).

The interaction of methanogens with other bacteria through interspecies H₂ transfer in the fermentation process allows methanogens to gain energy for their own growth, while the accumulation of H₂ and other intermediates is prevented, which benefits the growth of H₂-producing bacteria allowing further degradation of fibrous feed material (Hegarty and Gerdes 1998). Methanogens are hydrophobic and therefore stick to feed particles as well as onto the surface of protozoa. Tokura et al. (1997) observed that the number of methanogens associated with protozoa reached a maximum (10 to 100 times pre-feeding levels) after feeding, when the rate of fermentation is the highest. It was shown that the symbiotic relationship of methanogens and protozoa may generate 37% of rumen CH₄ emissions (Finlay et al. 1994).

Although methanogens are only directly involved in the very terminal stages of fermentation, they are very important because they are capable of effectively utilizing electrons in the form of H₂ to reduce CO₂ to CH₄, thereby maintaining low H₂ pressure in the rumen. Thus, in their absence, organic matter could not be degraded as effectively in the gut (McAllister et al. 1996). However, since CH₄ has no nutritional value to the animal, its production represents a loss of dietary energy to the animal. In general, CH₄ production in cattle constitutes about 2–12% of dietary GEI (Johnson and Johnson 1995). Reduction in CH₄ production can result from a decreased extent of fermentation in the rumen or from a shift in the VFA pattern towards more propionate and less acetate. Tamminga (1992) noted that if a decreased feed ruminal degradation is compensated for by an increased digestion in the small intestine instead of in the hindgut, it could be considered an advantage for the animal.

Measurement of Methane Emission from Dairy Cattle

Different methods used to measure CH_4 from animals have been reported in the literature. These include the use of respiration calorimetry chambers (Kelly et al. 1994; Galbraith et al. 1998; Murray et al. 1999), isotopic techniques (Murray et al. 1976; France et al. 1993), tracer techniques [sulfur hexafluoride (SF_6)], (Johnson et al. 1994; Kaharabata et al. 2000; Boadi and Wittenberg 2002) and mass balance/micro-meteorological techniques (Lassey et al. 1997; Sauer et al. 1998; Harper et al. 1999). The advantages and disadvantages of each method have been reviewed by Johnson and Johnson (1995). Examples of measured CH_4 emission values from dairy cattle are shown in Table 2.

Equations for predicting CH_4 emissions were developed mostly from data using the respiration calorimetry chamber to define the relationship between energy intake and CH_4 production, and are based mainly on the diet characteristics. The environment inside the respiration chamber is controlled and animals are under feed restriction during measurement. Therefore, data from the chamber cannot be applied under every farm situation, especially where animals are grazing and pasture quality is changing. Dynamic and mechanistic models to predict CH_4 from ruminants have also been established to simulate ruminal fermentation under a variety of nutritional conditions (Benchaa et al. 1998, 2001; Mills et al. 2001). Benchaa et al. (1998) showed that mechanistic models allow the prediction of CH_4 production more accurately than simple regression equations, under a large variation of diet composition. Regression analysis showed good agreement between observed and predicted results by modelling experimental data taken from the literature ($r^2 = 0.76$, root mean square prediction error = 15.4%; Mills et al. 2001). Although these models have usefulness in the prediction of CH_4 production from animals under the conditions from which the equations or models are developed, they are of limited use in the prediction of CH_4 production when intake is unknown or when the rumen is disturbed (Johnson et al. 2001).

Recent studies have been directed towards measurement of enteric CH_4 emissions under typical farm conditions in order to reflect existing feeding and management conditions. Variations can be seen in CH_4 emission measurements and efficiency of CH_4 production (L kg^{-1} milk) shown in Table 2. These can be attributed to differences in diet quality and quantities fed, animal body weight, level of milk production and also differences in methods used for estimating CH_4 emissions in each study. The studies of Kinsman et al. (1995), Sauer et al. (1998) and Kaharabata et al. (2000), which were conducted at the same dairy barn with similar animals and feeding regimes, show similar CH_4 emission efficiency values even though a different method of measuring CH_4 was used in the study of Kaharabata et al. (2000). There are limited studies on emission estimates of dairy cattle from other milking herds under different farm conditions in Canada. Thus, there is a need for measurement of values from other Canadian dairy herds to establish CH_4 emission values across the country. This will enable us to refine our overall estimates and to

properly evaluate mitigation strategies to reduce CH_4 emissions.

EXISTING ENTERIC METHANE MITIGATION STRATEGIES

Joblin (1999) stated that the management of H_2 production in the rumen is the most important factor to be considered when developing strategies to control ruminant CH_4 emissions. It should therefore be possible to reduce CH_4 production by inhibiting H_2 liberating reactions or by promoting alternative H_2 -using reactions or routes for disposing of H_2 during fermentation. Mitigation strategies that reduce ruminal methanogenesis by these mechanisms are discussed hereafter.

Improving Animal Productivity

In general, when animal productivity is improved through nutrition, management, reproduction or genetics, CH_4 production per unit of meat or milk is reduced. The amount of feed energy associated with animal maintenance is about 70-75% in beef cattle and 50% in dairy cattle (Mathison et al. 1998). The remaining feed energy is used for production. Thus as productivity increases, CH_4 emissions go up slightly, but CH_4 emissions per unit of product decrease (Johnson et al. 1996). This strategy is widely used in the dairy industry in Canada, where fewer animals are required to produce the same amount of milk, thereby reducing the number of animals maintained in production and the overall CH_4 produced.

Kirchgeßner et al. (1995) estimated that increasing milk production of dairy cows from 5000 to 10 000 L of milk annually in the EU, by using high grain rations or by improving the genetic merit of the dairy cow, would increase total CH_4 production per animal per year by 23% (i.e., from 110 to 135 kg yr^{-1}). However, CH_4 production per kg of milk produced would be reduced by 40% (i.e., from 0.022 to 0.014 $\text{kg of CH}_4 \text{ kg milk}^{-1}$). Therefore, overall CH_4 emissions could be decreased by reducing animal numbers while maintaining milk production. It was also reported that annual milk production for dairy cows in Queensland, Australia, increased from 2924 L head⁻¹ in 1988 to 4046 L head⁻¹ in 1996, through improvement in feed quality and management, and this was associated with a 26% reduction in CH_4 emissions per litre of milk (Howden and Reyenga 1999). However during this period, average feed intake and total CH_4 emissions (kg yr^{-1}) increased by 4 and 6%, respectively. These studies demonstrate that although the amount of CH_4 produced per animal per day increases, CH_4 produced per kg of milk declines. Improving productivity with the use of high grain diets must however be evaluated in terms of its cost of production and use of fertilizers and machinery, which will increase fossil fuel use and increase N_2O emissions. The cultivation and use of high-quality forages, which are cheaper than grain and do not involve increased use of fossil fuel through tillage, has been shown to be a sustainable option for producers and an efficient way to decrease N_2O emissions and increase soil carbon stores (Johnson et al. 1996).

Changes in milk pricing, from systems based on butterfat content to systems based on other components of milk, such

Table 2. Measured methane production from dairy cows

References	Technique	Animals	Milk (kg d ⁻¹)	CH ₄ (L cow ⁻¹ d ⁻¹)	CH ₄ (L kg of milk ⁻¹) ^a
Sechen et al. (1989)	Respiration calorimetry	6 lactating cows (BW = 603 kg)	37.1	557	15
Sechen et al. (1989)	Respiration calorimetry	6 lactating cows (BW = 603 kg)	41.3	470	11.4
Kirchgeßner et al. (1991)	Respiration calorimetry	67 lactating cows (BW = 583 kg)	17	420	24.7
Kinsman et al. (1995)	Mass balance	118 lactating cows (BW = 602 kg)	28.5	552	19.4
Sauer et al. (1998)	Mass balance	88–109 lactating cows (BW = 600 kg)	29	622	21.4
Kaharabata et al. (2000)	Atmospheric SF ₆ tracer	90 Holstein cows (BW = 600 kg)	28.5	542	19
Kaharabata et al. (2000)	Atmospheric SF ₆ tracer	147 dry heifers (BW = ND)	—	631	—
Westberg et al. (2001)	Internal SF ₆ tracer	4 lactating cows (BW = 673 kg)	22	623	28.3
Westberg et al. (2001)	Internal SF ₆ tracer	4 lactating cows (BW = 673 kg)	22	566	25.7
Johnson et al. (2002)	Room SF ₆ tracer	36 lactating cows (BW > 600 kg)	32.3	543	16.8
Johnson et al. (2002)	Room SF ₆ tracer	36 lactating cows (BW > 600 kg)	39.3	550	14
Johnson et al. (2002)	Room SF ₆ tracer	36 lactating cows (BW > 600 kg)	39.1	637	16.3

^aCalculated figures.

as protein, has been suggested to reduce CH₄ emissions (Johnson et al. 1996). Fat content of milk accounts for about 48% of the energy content of milk and therefore reducing milk fat content will decrease the need for feed energy, which, in turn, will reduce CH₄ production. A change in milk pricing based on solid-non-fat has been estimated to reduce CH₄ emissions from US milk cows by 15% (Johnson et al. 1996). With the demand for low fat milk increasing, pricing based on protein will encourage producers to modify feeding regimes to include the use of highly digestible protein feeds which will increase productivity and reduce CH₄ emissions. However high protein feeds are costly in dairy rations, and excessive amounts of nitrogen may be excreted in urine and feces. The impact on the environment as well as costs associated with such a strategy must be evaluated in terms of the overall benefits that can be achieved.

Increasing animal productivity is beneficial in reducing CH₄ production if animal numbers are reduced correspondingly. It can be concluded that continuous improvement of the efficiency of cattle production by nutritional and management strategies is a profitable and reduces CH₄ production.

Production Enhancing Agents (Bovine Somatotropin)

Production enhancing agents are available for use to increase production efficiency in cattle. Bovine somatotropin (bST) for dairy cows is a naturally occurring growth hormone produced by the pituitary gland. Recombinant bST, an identical molecule, is produced using biotechnology and has been shown to increase milk production in US dairy cows (Johnson et al. 1992). In general, the use of bST leads to an increase in milk production of 10–20%, and therefore animal numbers can be reduced to lower total enteric emissions (Johnson et al. 1996; Clemens and Ahlgrimm 2001). Johnson et al. (1996) estimated that the use of bST to improve US dairy cattle productivity could result in decreased CH₄ emissions (% of GEI) by about 9%. There is still a considerable debate regarding the health risks of bST, even though it has been approved for use in the United States. Consumers and health authorities are concerned about possible hormone residues in milk, which prevents its use in Canada.

Nutritional and Management Strategies

Type of Carbohydrates

The nature and rate of fermentation of carbohydrates influence the proportion of individual VFA formed and thus the amount of CH₄ produced. Fermentation of cell wall carbohydrates produces more CH₄ than fermentation of soluble sugars, which produce more CH₄ than fermentation of starch (Johnson et al. 1996). Diets rich in starch that favor propionate production will decrease CH₄ production per unit of fermentable organic matter in the rumen. Conversely, a roughage based diet will favor acetate production and increase CH₄ production per unit of fermentable organic matter (Johnson and Johnson 1995). Increasing the intake of concentrates from 40 to 68 g DM kg^{-0.75} decreased CH₄ production from 9.2 to 5.3% of GEI (McAllister et al. 1996). Very high grain diets (≥90%) such as the ones commonly fed in US feedlots may reduce CH₄ losses to 2–3% of GEI (Johnson and Johnson 1995). When highly digestible fiber sources like beetpulp were fed to steers at high levels of intake, CH₄ losses reported were 4–5% of GEI (Kujawa 1994). With mixed diets, the quantity of CH₄ produced will depend on the level and type of grain in the diet and the fermentation that results. Increasing barley supplementation from 0 to 75% (DM basis) in grass silage diets at high levels of intake (1.5 (maintenance) increased the volume of CH₄ produced per sheep per day, but decreased CH₄ per unit of digested feed organic matter by 8% (Moss et al. 1994).

Fermentation of structural carbohydrates results in a greater loss of GEI in the form of CH₄ than fermentation of soluble sugars and starches. This is a consequence of decreased rates of ruminal fermentation and passage out of the rumen, which favor a higher acetic:propionic acid ratio (Hegarty and Gerdes 1998). On the contrary, high grain diets fed at high intake levels are associated with high rates of ruminal digestion and passage which favor a higher propionic acid production (Hegarty and Gerdes 1998). High digestion rates of grains will also lower ruminal pH, which inhibits the growth of methanogens and protozoa (Hegarty 1999). Rumen fluid from cows that were fed 90% concentrate had lower ruminal pH values (6.22 vs. 6.86), higher VFA concentration (85 vs. 68 mM) and lower acetate:propionate ratio (2.24 vs. 4.12) than rumen fluid from cows fed

100% forages (Russell 1998). Russell (1998) noted that as much as 25% of the decrease in acetate:propionate ratio could be explained by the effect of pH alone. Cows with low milk yields usually receive higher roughage diets, which are replaced by higher concentrate diets as milk yield increases. As a consequence, CH₄ production is lower in the high producing dairy cow. Thus optimizing the diet to match the production level will reduce CH₄ production.

The increased use of grain to improve feed efficiencies and reduce CH₄ output per product will, however, be accompanied by the increased use of fossil fuel for chemical nitrogen fertilizer and machinery, which contributes nitrous oxide and fossil carbon to the GHG budget. These inputs must be evaluated in terms of the net contribution to total GHG to evaluate if there is a net reduction in GHG emission. Costs associated with grain feeding as opposed to forage feeding must also be evaluated when adopting such a strategy. Moreover, the decision to increase the utilization of grain in dairy cow rations to reduce CH₄ production should take into account the importance of ruminants in converting fibrous feeds, unsuitable for human consumption, to high-quality protein sources (i.e., milk and meat). Implementation of such strategies must therefore be balanced against the resultant reduction in CH₄.

Level of Intake

An increase in feeding level induces lower CH₄ losses as % of GEI. Johnson and Johnson (1995) noted that CH₄ loss as % of GEI declined by 1.6 percentage units for each multiple increase of intake. This is caused mainly by the rapid passage of feed out of the rumen (Mathison et al. 1998; Hegarty 2001a). As a result of the increased passage rate, the extent of microbial access to organic matter is decreased, which in turn reduces the extent and rate of ruminal dietary fermentation (Mathison et al. 1998). Okine et al. (1989) observed a 29% decrease in CH₄ production of cattle when the fractional passage rate of particulate matter was increased by 63% by placing weights in the rumen. About 28% of the variation in CH₄ production was attributed to the mean retention time. Also, a rapid passage rate favors propionate production, which is a competitive pathway for the use of H₂. However, the extent to which intake levels affect passage rate of roughages is proportionally less than with concentrate or mixed diets (Mathison et al. 1998). When mixed diets with less than 25% concentrates are fed, the effect of intake on passage rate of roughages is limited (Galyean and Owens 1991). Benchaar et al. (2001) reported an increase in CH₄ (Mcal d⁻¹) with increasing DMI, irrespectively of the diet type (100 alfalfa vs. 30% alfalfa +70% of concentrate mixture). However, when CH₄ was expressed relative to GEI or digestible energy, CH₄ yield decreased when DMI increased.

Forage Species and Maturity

CH₄ production in ruminants tends to increase with maturity of forage fed, and CH₄ yield from the ruminal fermentation of legume forages is generally lower than the yield from grass forages (McAllister et al 1996; Moss et al. 2000). When timothy hay was replaced with alfalfa in the predic-

tion model of Benchaar et al. (2001), CH₄ production (relative to digestible energy) decreased by 21%. The lower CH₄ loss observed with legumes compared to grasses can be attributed to the lower proportion of structural carbohydrates in legumes and faster rate of passage, which shift the fermentation pattern towards higher propionate production (Johnson and Johnson 1995).

In New Zealand, Friesian and Jersey dairy cows grazing sulla (*Hedysarum coronarium*), a condensed tannins-containing legume, emitted less CH₄ per unit of DMI (19.5 g CH₄ kg DM⁻¹) compared to cows grazing perennial ryegrass pasture (24.6 CH₄ kg DM⁻¹) (Woodward et al. 2002). The authors concluded that condensed tannins containing legumes such as sulla do effectively reduce CH₄ production from dairy cows without compromising milk solids production.

With regard to forage maturity, Robertson and Waghorn (2002) observed that CH₄ production from dairy cows grazing pastures in September (spring) was 4.5–5.3% of GEI, while it was increased to 6–7% of GEI when cows grazed in December (summer) of the same season. The authors stated that grass maturity accounted for the increased proportion of GEI lost as CH₄ from cows grazing over the lactation period. Selective grazing which is associated with grass maturity is also a factor to consider in the variation of CH₄ production.

Feeding Frequency

Low meal frequencies tend to increase propionate production; reduce acetic acid production and lower CH₄ production in dairy cows (Sutton et al. 1986). This effect is associated with the lowering of methanogens as a result of high fluctuations in ruminal pH, since low meal frequencies increase diurnal fluctuations in ruminal pH that can be inhibitory to methanogens (Sutton et al. 1986; Shabi et al. 1999). On the other hand, more frequent feeding was shown to increase the acetate: propionate ratio (Sutton et al. 1986; French and Kennelly 1990). French and Kennelly (1990) observed that when concentrates were fed to Holstein cows in 12 equal portions at 2-h intervals compared to two equal portions, the higher feeding frequency tended to elevate ruminal pH, increase acetate:propionate ratio and milk fat percentages. When lactating cows were supplemented with protein concentrate either two or five times daily, average rumen pH was higher and propionate lower in cows supplemented with five meals (Casper et al. 1999). Muller et al. (1980) observed that lactating cows produced 10% more CH₄ (% of GEI) in response to feeding concentrates six times daily compared to when they were fed twice daily. Producers are encouraged to increase their feeding frequency to reduce daily fluctuations in ruminal pH and to ensure efficient digestion and milk production. Thus, low frequency feeding as a strategy to reduce CH₄ production would not be practical to producers.

Forage Processing

Grinding or pelleting of forages to improve the utilization by ruminants has been shown to decrease CH₄ losses per unit of feed intake by 20–40% when fed at high intakes (Johnson et al. 1996). The lowered fiber digestibility,

decreased ruminally available organic matter and faster rate of passage associated with ground or pelleted forages can explain the decline in CH_4 production (LeLiboux and Peyraud 1999). These effects are more pronounced in animals offered feed at ad libitum level of intake compared to those whose intake is limited (LeLiboux and Peyraud 1999). Feeding ground diets instead of chopped diets reduced the molar proportion of acetic acid, and increased the molar proportions of propionic and valeric acids (LeLiboux and Peyraud 1999). However, fine grinding of forages has not proved to be economical to dairy producers because of increased incidence of acidosis and lower milk fat content of milk, which are associated with a lower effective fiber content of finely ground forages.

Forage Preservation

There is limited information with regard to the effects of forage preservation on CH_4 production. Methane production (% of GEI) was shown to be lower when forages were ensiled than when dried (Sundstol 1981). This is because digestion is reduced in the rumen with ensiled forages due to the extensive fermentation that occurs during silage making. Shingfield et al. (2002) observed that rumen fermentation of grass-silage-based diets was characterized by a higher molar proportion of butyrate and a lower proportion of acetate compared to hay-based diets. Total CH_4 production (Mcal d^{-1}) was depressed by 33% by the utilization of alfalfa silage instead of alfalfa hay, using a mechanistic model approach to predict CH_4 emissions from ruminants (Benchaa et al. 2001). Kirkpatrick and Steen (1999) observed no differences between forages conserved as silage vs. forages conserved by freezing (directly after harvesting) on CH_4 energy loss (% of GEI). Silage additives such as bacterial inoculants and organic acids are used to enhance the quality and palatability. Shingfield et al. (2002) also observed that the addition of inoculant-enzyme preparation during ensiling lowered acetic acid and increased propionate production as well as branched chained VFA in the rumen when compared to formic acid. Thus, the addition of inoculant-enzyme during silage making would seem to hold a greater potential for reducing enteric CH_4 emissions than the addition of formic acid.

Grazing Management

Implementing proper grazing management practices to improve the quality of pastures will increase animal productivity and lower CH_4 per unit of product. McCaughey et al. (1997) observed that CH_4 production was greatest for steers continuously grazing at low stocking rates ($1.1 \text{ steer ha}^{-1}$; 306.7 L d^{-1}) and least for steers grazing continuously at high stocking rates ($2.2 \text{ steers ha}^{-1}$; 242.2 L d^{-1}). At higher stocking rates, forage availability and intake are low. When pastures were rotationally grazed, stocking rates had no effect on CH_4 production (L d^{-1}) (McCaughy et al. 1997). At low stocking rates ($1.1 \text{ steer ha}^{-1}$), CH_4 production ($\text{L ha}^{-1} \text{ d}^{-1}$) was 9% lower on rotational grazing than continuous grazing (McCaughy et al. 1997). Measurements of CH_4 production from grazing beef cows indicated a 25% reduction in CH_4 losses with alfalfa-grass pastures (7.1% of GEI) compared to grass-only pastures (9.5% of GEI) (McCaughy et al.

1999). Early grazing of alfalfa-grass pastures, reduced CH_4 production (% GEI) by 29–45% in steers compared to grazing at mid and late seasons (Boadi et al. 2002).

There is evidence that pasture-based dairy farming systems can be as profitable as confinement systems (Dartt et al. 1999; White et al. 2002). A major drawback of intensive grazing is the likelihood of achieving slightly lower milk production than with confinement feeding, however net returns under managed intensive grazing are greater because of lower feeding costs associated with pasture forages (Parker et al. 1992; Hanson et al. 1998; Dartt et al. 1999). The use of pasture reduces the need for growing, harvesting, and preserving forages on farm. This means less machinery, fuel, and fertilizers, and therefore less GHG produced with grazing systems.

Harper et al. (1999) observed that CH_4 production from heifers grazing pastures was $321.2 \text{ L head}^{-1} \text{ d}^{-1}$, which was 7.7–8.4% of GEI. When the same animals were put into the feedlot and fed a high grain diet, they produced $98 \text{ L head}^{-1} \text{ d}^{-1}$ (1.9–2.2% of GEI). On the other hand, Robertson and Waghorn (2002) observed that CH_4 loss (% of GEI) was similar from Friesian and Holstein dairy cows (150 d post calving) fed either a TMR (5.7%) or grazing pastures (6.3%). The discrepancy between these two studies is due to differences in the proportion of highly digestible grain used in the diet for feedlot heifers (80% of dietary DM) compared to that used in TMR fed to dairy cows (50% of dietary DM). Dhiman et al. (2001) observed that CH_4 emissions ($\text{g cow}^{-1} \text{ d}^{-1}$, g kg^{-1} of DM intake or g kg^{-1} of fat-corrected milk) in lactating cows fed conserved forages and grain-based diets was similar to those fed high-quality pastures only. Pasture quality becomes a critical factor in ensuring lower CH_4 emissions from grazing animals. Robertson and Waghorn (2002) confirmed this when they observed that at 240 d post calving (i.e., March), cows grazing pastures had significantly higher CH_4 emissions (7% of GEI) compared to those receiving a TMR (6.3% of GEI), while there was no difference between the two feeding regimen at 150 d post calving (i.e., December). In summary, when comparing CH_4 emissions from grazing vs. feedlot systems, factors such as pasture quality, level of concentrate provided in the feedlot as well as operational costs have to be taken into account.

Manipulation of Rumen Fermentation

Addition of Fats

Fats are added to dairy cattle diets to increase the energy density of diets, enhance milk yield and modify the fatty acid composition of milk fat (Murphy et al. 1995; Ashes et al. 1997). Fats have also been shown to depress CH_4 production (Dong et al. 1997a; Machmuller and Kreuzer 1999; Dohme et al. 2000). It has been shown that the medium chain fatty acids (C_8 – C_{16}) cause the greatest reduction in CH_4 production (Dohme et al. 2000). Dohme et al. (2000) observed from in vitro studies that with the addition of 53 g kg^{-1} DM of palm kernel oil, coconut oil, or canola oil, CH_4 production was reduced by 34, 21 or 20%, respectively. Machmuller and Kreuzer (1999) observed that when coconut oil was fed to sheep at dietary levels of 3.5 and 7%,

daily CH_4 production was reduced by 28 and 73%, respectively. Mathison (1997) reported that CH_4 production was reduced by 33 when 4% of canola oil was added to a diet containing 85% concentrate in a feedlot study. On the other hand, Johnson et al. (2002) recently observed that supplementation of diets with oilseeds at 2.3, 4 or 5.6% of the diet did not affect CH_4 production, but tended to increase the efficiency of milk produced per unit of CH_4 produced especially at the 4% supplementation. Differences in the response of CH_4 production following the addition of fats may be due to the type of fat added and how available it is in the rumen. Dohme et al. (2000) observed no differences in CH_4 production when protected fats such as prilled fat (6.9 mmol d^{-1}), crystalline fat (6.5 mmol d^{-1}) and Ca soap fats (6.6 mmol d^{-1}) were supplemented in vitro compared to control diets (7.0 mmol d^{-1}). Thus when protected fats are used, there is no observed reduction in methanogenesis.

The major problem associated with excessive fat addition (more than 5-6% of the ration) is that it depresses fiber degradation in the rumen (Dong et al. 1997a; Mathison et al. 1998), and reduces acetate production and milk fat content (Jenkins et al. 1996; Ashes et al. 1997). Mathison (1997) observed that digestibility of fiber was reduced with the addition of canola oil, such that the metabolizable energy content of the diet was not different from that of the control barley-based diet. Dong et al. (1997a) observed that of the three oils (coconut, canola, and cod-liver oil) used in an artificial rumen system, coconut oil depressed fiber digestion the most. Although the addition of fats appears to reduce CH_4 production, excessive addition would not result in increased energy availability to the animal due to depressed fiber digestion. Also, saturated fatty acids such as coconut oil have been shown to have a toxic effect on protozoa and methanogens (Dong et al. 1997a; Machmuller and Kreuzer 1999). Feeding 3.5 and 7% coconut oil reduced ciliate protozoa to $1.0 \times 10^5 \text{ mL}^{-1}$ and $0.3 \times 10^5 \text{ mL}^{-1}$, respectively, compared to the controls ($1.7 \times 10^5 \text{ mL}^{-1}$) (Machmuller and Kreuzer 1999).

The depression in CH_4 production with addition of unsaturated fats has been attributed to the fact that these fatty acids can serve as electron acceptors during biohydrogenation in the rumen (McAllister et al. 1996; Hegarty 1999). Long-chain fatty acids are non-fermentable and therefore may decrease the percentage of CH_4 that can be produced (Johnson and Johnson 1995). Van der Honing et al. (1981) observed a 10-15% decrease in CH_4 loss from dairy cows fed 5% tallow or soybean oil. This reduction in CH_4 production was attributed to a decrease in fermentable substrate rather than to a direct effect on methanogenesis.

Ionophores

Ionophores are highly lipophilic substances, which are able to shield and delocalize the charge of ions and facilitate their movement across membranes (Mathison et al. 1998). Monensin is the most commonly used and studied ionophore, with others such as lasalocid, tetronasin, lysocellin, narasin, salinomycin and laidomycin also being used commercially. Ionophores, which are added to ruminant diets to improve the efficiency of feed utilization, have

been shown to decrease CH_4 production (Mathison et al. 1998; Moss et al. 2000). Monensin addition to the diets of 109 milking Holsteins at 24 ppm, decreased CH_4 production ($\text{L d}^{-1} \text{ cow}^{-1}$) by 21% from control levels (Sauer et al. 1998). Feed intake decreased from $16.1 \text{ kg of DM d}^{-1} \text{ cow}^{-1}$ to $14.5 \text{ kg of DM d}^{-1} \text{ cow}^{-1}$, while milk production increased from $27.6 \text{ kg d}^{-1} \text{ cow}^{-1}$ to $31.5 \text{ kg d}^{-1} \text{ cow}^{-1}$ (Sauer et al. 1998). The observed increase in propionate production and decrease in CH_4 production that accompany monensin feeding have been associated with the selective reduction of Gram-positive ruminococci, and the proliferation of the Gram-negative bacteria with a concurrent shift in the fermentation from acetate to propionate production (Newbold et al. 1988; Van Nevel and Demeyer 1995).

Cattle studies have shown that CH_4 production was not suppressed with prolonged or repeated use of ionophores (Van Nevel and Demeyer 1995; McCaughey et al. 1997; Sauer et al. 1998). In grain-fed sheep with repeated monensin addition, Mbanzamihigo et al. (1996) showed that CH_4 production rates remained lower than that of the controls, for at least 35 d. The poor persistence of monensin in suppressing CH_4 activity may be due to the development of resistance to the antibiotic. Methanogens have not been screened for development of resistance to monensin (Hegarty 2001b). Strains of rumen methanogens do, however, differ in susceptibility to monensin (Chen and Wolin 1979) and prolonged use of antibiotic may simply select for non-susceptible strains. Because of the beneficial effects of ionophores on feed efficiency and productivity, the use of ionophores could be an economically justifiable option. However, with the rising public and health authorities concerns about the use of chemical or antibiotic feed additives in animal production, the use of ionophores to mitigate CH_4 production might not be the best conceivable option.

Defaunation

Defaunation, which is the elimination of protozoa from the rumen by dietary or chemical agents, has been shown to reduce ruminal CH_4 production by about 20 to 50% depending on the diet composition (Whitelaw et al. 1984; Itabashi et al. 1994; Van Nevel and Demeyer 1996). Whitelaw et al. (1984) observed that faunated cattle fed barley diets at restricted levels lost about 12% of GEI as CH_4 compared to 6-8% of GEI in ciliate-free animals. Protozoa in the rumen are associated with a high proportion of H_2 production, and are closely associated with methanogens by providing a habitat for up to 20% of rumen methanogens (Stumm et al. 1982; Newbold et al. 1995a). Finlay et al. (1994) reported that protozoa could account for 37% of the total CH_4 production. It is assumed that there is a symbiotic H_2 transfer between anaerobic protozoa and methanogens (Stumm et al. 1982; Ushida and Jouany 1996). The reduced ruminal methanogenesis observed with defaunation can be attributed to factors such as a shift of digestion from the rumen to the hind gut (Van Nevel and Demeyer 1996) or the loss of methanogens associated with protozoa during defaunation (Hegarty 1999).

It has been shown that defaunation may depress fiber digestion, thus complete elimination of protozoa (rather

than selective defaunation) is not recommended as a method for reducing CH_4 (Itabashi 2001). On the other hand, protozoa have been reported to negatively affect ruminal protein metabolism through predation of bacteria, which reduces the flow of microbial protein leaving the rumen (Ivan et al. 2000; Koenig et al. 2000). Therefore, the use of defaunation to mitigate CH_4 production from ruminants should be weighed against its possible impact on the efficiency of the whole ruminal system. Defaunating agents or protozoal inhibitors are not currently available for commercial or practical use as many of the defaunating agents are toxic to the animal. The control of protozoa is unlikely to lead to H_2 accumulation or inhibition of fermentation, therefore it represents a promising method of CH_4 reduction. Further work is needed in this area to develop commercial means of controlling rumen protozoa (Klieve and Hegarty 1999).

Direct Inhibition by Chemicals

Bromoethanesulphonate (BES) is a potent inhibitor of methanogenesis because it is a structural analogue of the co-factor mercaptoethanesulfonic acid (coenzyme M) used by methanogenic bacteria (Mathison et al. 1998). Given that coenzyme M is found only in methanogens, BES works as a specific inhibitor when used in mixed microbial systems. Dong et al. (1997b) observed that BES depressed CH_4 production by 71% without significantly affecting organic matter digestibility and VFA concentrations in the artificial rumen (Rusitec). The reduction in CH_4 was attributed to the toxic effect of BES to methanogenic bacteria. However, a study in sheep showed that even though BES effectively depressed CH_4 production, its effectiveness persisted for only 3 d (Immig et al. 1995), suggesting that adaptation of the methanogenic population occurred. It can be concluded that while BES is an effective inhibitor of CH_4 production, it may not offer a viable method of reducing CH_4 on a commercial level.

Chlorinated CH_4 analogues (chloroform, carbon tetrachloride, methylene chloride) and related compounds such as trichloroethyl adipate and pyromellitic diimide have been reported to inhibit methanogenesis in a review by Van Nevel and Demeyer (1996). Because CH_4 analogues are directly toxic to methanogenic bacteria, their use results in decreased molar proportions of acetate and an increase in propionate. However, microbial populations in vivo have been found to adapt to or degrade many of these compounds so that favorable long lasting effects on animal performance and health are limited (Van Nevel and Demeyer 1996). In a recent study, the feeding of 500 ppm of 9,10-anthraquinone resulted in a decrease in CH_4 concentration throughout the 19 d feeding period in sheep (Kung et al. 2003). There was no indication of ruminal adaptation throughout the feeding period and feeding of 9,10-anthraquinone was not toxic to the animals (Kung et al. 2003). This was the first study to show that 9,10-anthraquinone can partially inhibit in vivo rumen methanogenesis. However its long-term effects must be evaluated. McCrabb et al. (1997) studied the effect of a chemical complex of bromo-chloromethane and cyclodextrin, which is chemically stable when added to feed, and reported that these two compounds as a complex reduced

CH_4 production without any negative effect on fiber digestion. Longer lasting effects of chemical agents may offer an avenue for mitigating enteric CH_4 production, but public health concern about chemical residues in milk will have to be addressed.

NEW POTENTIAL MITIGATION OPTIONS

Probiotics

There is very little information on the effects of probiotics on CH_4 production in dairy cattle. The effects of the most widely used microbial feed additives, *Saccharomyces cerevisiae* and *Aspergillus oryzae*, on rumen fermentation were earlier studied in vitro (Frumholtz et al. 1989; Mutsvangwa et al. 1992). *Aspergillus oryzae* was shown to reduce CH_4 by 50% as a result of a reduction in the protozoal population (Frumholtz et al. 1989). The addition of *Saccharomyces cerevisiae* reduced CH_4 by 10% in vitro, but was not sustained over a long period (Mutsvangwa et al. 1992). It has been shown that yeast culture influenced microbial metabolism and improved DMI, fiber digestion, and milk production in lactating cattle (Shaver and Garrett 1995; Doreau and Jouany 1998; Dann et al. 2000). However, the specific mode of action is still unknown. It has been proposed that probiotics provide nutrients, including metabolic intermediates and vitamins, that stimulate the growth of ruminal bacteria, resulting in increased bacterial population (Newbold et al. 1996). Another theory indicates that probiotics stimulate lactic-acid-utilizing bacteria, resulting in a reduction of lactic acid and a more stable ruminal environment. A less acidic ruminal environment favors the growth of cellulolytic bacteria, which in turn improves fiber digestion, feed intake, and production response (Williams et al. 1991; Yoon and Stern 1996).

Miller-Webster et al. (2002) recently showed that the inclusion of yeast culture products (YC1, Diamond-V XP, and YC2, A-Max) in a continuous culture system increased DM digestion and propionic acid production whereas it reduced acetic acid production and protein digestion compared with the control. Eun et al. (2003) reported that brewers yeast culture enhanced the activity of bacteria that convert H_2 to acetate and decreased CH_4 output by 25% in a continuous culture of ruminal microorganisms. In a previous study, Chiquette and Benchaar (1998) reported no effect on molar proportions of ruminal VFA when a mixture of *Saccharomyces cerevisiae* and *Aspergillus oryzae* was added to the diet of dairy heifers. The effects of probiotics on fermentation pattern are not consistent across experiments and between strains of yeast (Newbold et al. 1995b). Doreau and Jouany (1998) found no effect of *Saccharomyces cerevisiae* on fermentation in lactating dairy cows, while Takahashi et al. (1997) observed that a probiotic preparation significantly increased (+ 18%) CH_4 production in sheep. Although microbial preparations are commercially available as ruminant feed additives, there is a need for further research to establish the potential of probiotics for reducing CH_4 production in vivo. Producers are skeptical about the benefits of probiotics and there is a need to identify the dietary and management situations in which

probiotics can give consistent production benefits as well as the added effect of reducing CH₄ emissions (Moss et al. 2000).

Bacteriocins

Direct suppression of methanogens may be possible through stimulation of natural or introduced ruminal organisms to produce bacteriocins as a means of biological control (Klieve and Hegarty 1999). Bacteriocins are bacteriocidal compounds that are peptide or protein in nature, and are produced by bacteria. However, little information is available concerning their effect on methanogenesis. They often display a high degree of target organism specificity, although many have a very wide spectrum of activity (Kalmokoff et al. 1996). Nisin, an exogenous bacteriocin produced by *Lactococcus lactis*, is the best studied and understood bacteriocin. It has similar actions to monensin and is widely used in the food industry as a preservative in controlling food borne pathogens (Lee et al. 2002). In vitro, nisin stimulated propionate production, increased the ratio of propionate to acetate and reduced methanogenesis by 36% (Callaway et al. 1997). However, recent work indicated that some ruminal bacteria become nisin-resistant (Mantovani and Russell 2001) and an in vivo feeding trial indicated that nisin could not decrease the acetate: propionate ratio as observed with cattle consuming the same amount of monensin (350 mg d⁻¹) (Russell and Mantovani 2002). This suggests that nisin was either being degraded or the bacteria were becoming nisin-resistant. The HC5 bovicin bacteriocin from *Streptococcus bovis* (*S. bovis*) has also been shown to inhibit CH₄ by as much as 50% (Lee et al. 2002). Although exogenous bacteriocins may be safe and can be incorporated into feed, a limitation may be the degree of stability of these peptides in the ruminal environment, as rapid degradation by proteolytic enzymes could reduce their effectiveness (Klieve and Hegarty 1999).

Endogenous bacteriocins have been identified in the rumen (Kalmokoff et al. 1996; Teather and Forster 1998). A survey of 50 strains of *Butyrivibrio* spp. isolated from a variety of sources (sheep, deer and cattle) for bacteriocin production indicated a high incidence of bacteriocin-like activity (50%) (Kalmokoff et al. 1996). Although the potential for ruminally produced bacteriocins to suppress methanogens is unknown, their potential to improve ruminant production and modify microbial populations has been suggested by Teather and Forster (1998). Bacteriocins may therefore provide an alternative to ionophore antibiotics for manipulation of ruminal microbial populations. They have advantages over other antibiotics in terms of target specificity, broad spectrum of activity, and possibility of genetic transfer and manipulation into other organisms (Kalmokoff et al. 1996). Bacteriocins could possibly be delivered as microbial inoculants for in situ production of the bacteriocin in the rumen or in silage (Kalmokoff et al. 1996). Given the fact that *S. bovis* produces a very potent bacteriocin (bovicin HC5), which reduces methanogenesis (Lee et al. 2002), silage fermentation can be a vehicle for delivering bacteriocins to the rumen (Kalmokoff et al. 1996). Controlled colonization of the rumen by genetically engineered ruminal

bacteria is a great challenge (Teather and Forster 1998). Also, there is a need to develop rapid and accurate techniques to characterize the existing ruminal populations in terms of bacteriocin production and resistance. Efforts are under way to clone bacteriocin genes and develop DNA probes for the detection of these genes in rumen samples (Teather and Forster 1998). Currently, the genomes of several lactic acid bacteria that produce bacteriocins have been sequenced (Konings et al. 2000). These organisms have found wide application in the manufacturing of fermented foods and drug industry. Recent progress has been made in the construction of genetically modified lactic acid bacteria used in food products (Konings et al. 2000). It can be concluded that bacteriocins have the potential to reduce CH₄ production, but further studies in vivo are needed to establish their adaptability and long term effectiveness as a feed additive.

Archaeal Viruses

Another possible method of biological control of methanogens is the use of archaeal viruses (bacteriophages) (Klieve and Hegarty 1999). Bacteriophages are obligate pathogens that can infect and lyse bacteria and methanogens. They are highly host-specific. Although the presence of bacteriophages in the rumen is well known (Klieve and Bauchop 1988; Mackie and White 1990), knowledge of archaeal viruses is still limited. No viruses have been recorded from the rumen archaea and few from methanogens in general (Klieve and Hegarty 1999). Taraknov (1994) reported the successful use of *S. bovis* phage to suppress amylolytic bacteria. Klieve and Hegarty (1999) noted that without a considerable increase in knowledge of the genetic diversity and viral susceptibility of methanogens and the host range of archaeal viruses, it is difficult to assess the potential of archaeal viruses as a biological control agent.

Immunization

In the past 3 yr, researchers in Australia have vaccinated sheep with a number of experimental vaccine preparations against methanogens, so that the animals produce antibodies to methanogens (<http://www.csiro.au>). Methane production was reduced between 11 and 23% in vaccinated animals and productivity was improved. No long- or short-term adverse effects on sheep were found. Researchers anticipate that commercial vaccines will allow a 3% gain in animal productivity and a 20% reduction in CH₄ production (<http://www.csiro.au>). It is important to note that the vaccines currently under development are based on cultivable methanogens. However, the work of Whitford et al. (2001) showed that most ruminal methanogens have not yet been cultivated. Hegarty (2001b) noted that vaccine preparations are likely to work on some methanogens and not on others; thus, monitoring and assessment of efficacy will be required for novel control measures such as vaccines.

Reductive Acetogenesis

A technology that may hold some promise in the long-term of diverting electrons from methanogens is the production

of acetic acid by acetogens (Joblin 1999). In the gut of termites and rodents, acetogens convert excess H_2 to acetic acid, which is then utilized by the host (Joblin 1999). However, in the rumen, acetogens are few and cannot compete effectively with methanogens for H_2 ions, because they have a lower affinity for H_2 than methanogens (Greening and Leedle 1989; Nollet et al. 1998).

Carbon flux studies in the rumen of sheep revealed that rumen acetogenesis occurs in the first 24 h after birth, but is subsequently displaced by methanogenesis (Morvan et al. 1994); methanogens easily out-compete the acetogens for the low concentration of H_2 normally encountered in the rumen (Joblin 1999). Thus, methanogens have to be inhibited to allow H_2 pressure to rise before acetogenesis can be significant as an alternate H_2 sink in the rumen. Increasing the populations of acetogens through exogenous inoculations into the rumen could be useful for competing against methanogens (Joblin 1999). However, previous attempts at inducing acetic acid by inoculation with acetogens were not successful (Immig et al. 1996; Nollet et al. 1998).

Methane Oxidizers

CH_4 oxidizing bacteria have been isolated from different environments, including the rumen (Stock et al. 1964; Hanson et al. 1992; Moss et al. 2000). In vitro studies with stable carbon isotopes suggest that the extent of CH_4 oxidation to CO_2 is quantitatively minor (0.3 to 8%) in the rumen (Valdez et al. 1996; Kajikawa and Newbold 2000). Valdez et al. (1996) isolated a CH_4 oxidizing bacterium from the gut of young pigs, which decreased CH_4 accumulation when added to rumen fluid in vitro. However, this approach has not been validated in vivo. In the long-term, CH_4 oxidizers from gut sources could be screened for their activity in the rumen to reduce the proportion of ruminal gas in the form of CH_4 .

Propionate Enhancers

As a result of the growing awareness of the threat of microbial resistance to antibiotics, there is an increasing interest in alternatives to antibiotics as growth promoters (Moss et al. 2000). Dicarboxylic acids such as fumaric and malic acids have been studied in vitro as feed additives in ruminant diets (Callaway and Martin 1996; Lopez et al. 1999; Asanuma et al. 1999). Fumaric acid is an intermediate in the propionic acid pathway, in which it is reduced to succinic acid. In this reaction, H_2 ions are needed and therefore reducing fumaric acid may provide an alternative electron sink for H_2 . It was found that the addition of up to 500 μ mol of sodium fumarate in vitro decreased CH_4 production by 6% and increased DM digestibility of the basal diet by 6% after 48 h incubation (Lopez et al. 1999). Asanuma et al. (1999) showed that the addition of 20 mM of fumarate to cultures that were fermenting hay powder and concentrate incubated for 6 h significantly decreased CH_4 production by 5% and increased propionate production by 56%, while with the addition of 30 mM of fumarate, CH_4 declined by 11%, and propionate production increased by 58% compared to the control. Their data suggested that most of the fumarate consumed was metabolized to propionate with little production of acetate and succinate, whereas a much larger amount of

succinate accumulated with the addition of 30 mM of fumarate. However, when incubation time was prolonged to 12 h, most of the succinate was metabolized to propionate.

There is little information available on the actual effects of fumaric acid on fermentation and animal performance in vivo. Isobe and Shibata (1993) observed that the proportion of acetic acid and propionic acid increased following the addition of fumaric acid whereas the proportion of the higher acids decreased. The effects of salinomycin (15 ppm) plus fumaric acid (2%) supplemented to diets of Holstein steers increased the molar proportion of propionic acid and decreased CH_4 production ($L\ kg\ DMI^{-1}$) by 16% and had no effect on DM digestibility (Itabashi et al. 2000). Bayaru et al. (2001) found that CH_4 production was reduced by 23% when fumaric acid added to sorghum silage was fed to Holstein steers. The authors observed that the addition of fumaric acid increased propionic acid formation and had no effect on DM digestibility. Fumaric acid was also shown to increase concentration of plasma glucose and milk protein synthesis in dairy cows due to an increase in propionic acid production (Itabashi 2001). The authors concluded that fumaric acid may be put to practical use for ruminant diets since it has the dual benefit of decreasing CH_4 production and increasing net energy retention. Malate, which is converted to propionate via fumarate, also increased propionate production and inhibited CH_4 production in vitro (Martin et al. 1999). However, malate failed to increase ruminal propionate concentrations in feedlot cattle and did not affect CH_4 production (Martin et al. 1999; Montano et al. 1999) although it stimulated daily gains in steers (Martin et al. 1999). There is a need for further testing and evaluation of these enhancers in vivo to assess their potential as feed additives in the industry.

Essential Oils

There is an increasing interest in exploiting natural products as feed additives to manipulate enteric fermentation and possibly reduce CH_4 emissions from livestock production (Wallace et al. 2002; Wenk 2003). Essential oils are a group of plant secondary compounds that hold promise as natural additives for ruminants (Wallace et al. 2002). Essential oils are any of a class of steam volatile oils or organic-solvent extracts of plants (e.g., thyme, mint, oregano, sage) possessing the odor and other characteristic properties of the plant (mainly antimicrobial), used chiefly in the manufacture of perfumes, flavors, food preservatives, and pharmaceuticals (Wenk 2003). Essential oils are present in many plants and may play a protective role against bacterial, fungal, or insect attack. The antimicrobial activity of essential oils can be attributed to a number of small terpenoids and phenolic compounds, e.g. monoterpenes, limonene, thymol, carvacrol (Helander et al. 1998; Hammer et al. 1999; Wallace et al. 2002). The specific mode of action of essential oil constituents remains poorly characterized or understood (Helander et al. 1998).

The antimicrobial properties of essential oils have been shown through in vitro and in vivo studies to inhibit a number of bacteria and yeasts and to control fermentation gases, VFA, livestock waste odors and human pathogenic bacteria

Table 3. Summary of methane mitigation strategies for dairy cows

Strategy	Potential CH ₄ reduction	Technology availability/feasibility	Cost/production benefits
Improving animal productivity	20–30%	Feasible and practical	Increased feed cost increased milk production use of fewer animals less feed per kg of milk Increased feed intake Increased feed cost, machinery/fertilizer use increased milk production Increased cost of processing improved feed efficiency increased milk production Increased feed efficiency increased milk production Increased cost of fencing increased management of animals increased feed intake increased milk production Cheaper feed cost may need supplements reduced milk fat/protein content higher net return Increased feed intake increased milk production
Increasing concentrate levels at high levels of intake	25% or more	Feasible, for high producing cows, but may increase N ₂ O and CO ₂ emissions	
Processing of forages, grinding/pelleting	20–40%	Feasible	
Forage species and maturity	20–25%	Feasible	
Rotational grazing of animals/early grazing	9% or more	Feasible	
Managed intensive grazing vs. confined feeding		Feasible needs more investigation	
Use of high-quality forages/pastures	25% or more	Feasible	
Preservation of forage as silage vs. hay /additives Addition of fats	up to 33% (model prediction) up to 33%	Feasible Feasible and practical, but usage limited to 5–6% in diet Feasible, but not long lasting public concerns	Limited studies Increased cost of diet increased or no effect on milk production may or may not affect milk fat Increased feed efficiency decreased feed intake increased milk production May increase feed intake may increase milk production or no change Not quantified Reduced feed cost Practicability and cost to be assessed
Use of ionophores, e.g., monensin, lasolocid	11–30%	Feasible, needs more investigation	Economic feasibility ruminant adaptation and level of inclusion need to be evaluated Needs further investigation Production effects are to be evaluated
Use of probiotics	10–50% (<i>in vitro</i>)	Feasible, needs more investigation	Increased cost of chemicals production effects not established
Use of essential oils Use of bovine somatotropin (bST) Protozoa inhibitors	8–14% (<i>in vitro</i>) 9–16% 20–50% (<i>in vitro</i> and <i>in vivo</i>)	Feasible, needs more investigation Not approved for use in Canada Not available for practical use Possible microbial adaptation to fumaric acid Not available, needs more investigation	May increase cost of production increased gain Decreased feed intake increased feed efficiency
Propionate enhancers (fumarate, malate)	5–11% (<i>in vitro</i>) up to 23% (<i>in vivo</i>)	Not available, needs more investigation	
Use of acetogens	not quantified	May provide alternatives to ionophores needs more investigation No compounds registered for use No long lasting effects identified	
Use of bacteriocins, e.g., Nisin, bovicin HC5	up to 50% (<i>in vitro</i>)		
Use of methane inhibitors, e.g., BES, 9,10-anthraquinone	up to 71% (<i>in vitro</i>)		
Immunization	11–23%	Not available, needs more investigation	
Genetic selection (Use of high Net Feed Efficiency animals)	21%	Long term feasibility	

such as *Escherichia coli* 0157:H7, *Enterococcus faecalis* and *Salmonella* sp. (Helander et al. 1998; Hammer et al. 1999; Broudiscou et al. 2000; Wallace et al. 2002). For the purposes of controlling ruminal fermentation and CH₄ production, the effect of adding 0, 1 and 10% essential oil to 0.5 g of ground tall fescue and concentrate in the ratio of 2:8 or 8:2 was examined on in vitro gas production and fermentation by Lee and Ha (2002). The authors showed that supplementing 10% of essential oil increased ruminal pH and lowered NH₃-N, VFA concentration and cumulative CH₄ production over 48 h of incubation, when compared with the 0 or 1 % levels. There was no effect on CH₄ production following the addition of 1% essential oil to both substrates (Lee and Ha 2002). Broudiscou et al. (2000) screened 13 plant extracts for their action on fermentation in vitro and observed that protozoa numbers were little affected. On the other hand, methanogenesis decreased by 8.2% with *Salvia officinalis* and by 14.2% with *Equisetum arvense*, while it increased by 13.7% with *Lavandula officinalis* and 7.7% with *Solidago virgaurea*, indicative of diverse modes of action among plant extracts.

When sheep diets (60:40 silage:concentrate) were supplemented with 100 mg of essential oils head⁻¹ d⁻¹, Wallace et al. (2002) reported no effects on the ruminal concentration of VFA and protozoa numbers. Recently, Benchaar et al. (2003) did not observe any effects of dietary addition of essential oils on VFA concentrations, acetate:propionate ratio, or rumen microbial counts in lactating cows. The potential of essential oils for modulating ruminal function on a long-term basis has not been evaluated. It is also important to know the most effective level of inclusion of essential oils in the diet, as well as the possible adaptation of ruminal microorganisms to this feed additive.

Genetic Selection

Robertson and Waghorn (2002) observed that Dutch/US cross Holstein cows produced 8–11% less CH₄ (% of GEI) than New Zealand Friesian cows for about 150 d post calving, either when grazing or receiving a TMR. Hegarty (2001a) noted that the natural variation among animals in the quantity of feed eaten per unit of liveweight gain can be exploited to breed animals that consume less feed than the unselected population while achieving a desired rate of growth. Accordingly, to exploit such traits, the concept of Residual (Net) Feed Intake (RFI) was developed and used (Koch et al. 1963; Archer et al. 1998; Okine et al. 2001; Basarab et al. 2003). The RFI is moderately heritable ($h^2 = 0.39$), and is independent of the rate of gain (Archer et al. 2001). Okine et al. (2002) calculated annual CH₄ emissions from Canadian high NFE steers to be 21% lower than that for low NFE steers. Selection for high NFE in beef cattle also decreased manure N, P, K output due to a reduction in daily feed intake and more efficient use of feed, without any compromise in growth performance (Okine et al. 2002). The mean retention time of digesta has also been shown to be selectable among animals (Hegarty 2001b). Selecting animals for a faster passage rate of feed from the rumen would reduce CH₄ emissions per unit of food ingested. Faster passage rate of feed also affects propionate and

microbial yield; thus, selection of animals for this would also have major production benefits. Selecting animals with high NFE offers an opportunity to reduce daily CH₄ emissions without reducing livestock numbers.

CONCLUSIONS

Mitigation of CH₄ emissions can be effectively achieved by strategies that improve the efficiency of animal production, reduce feed fermented per unit of product, or change the fermentation pattern in the rumen (Table 3). Many current and potential mitigation strategies have been evaluated, but not all of them can be applied at the farm level, and in many cases the potential negative effects and associated costs have not been fully researched. Strategies that are cost effective, improve productivity, and have no potential negative effects on livestock production hold a greater chance of being adopted by producers.

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